بسم الله الرحمن الرحيم

THE NEWBORN INFANT

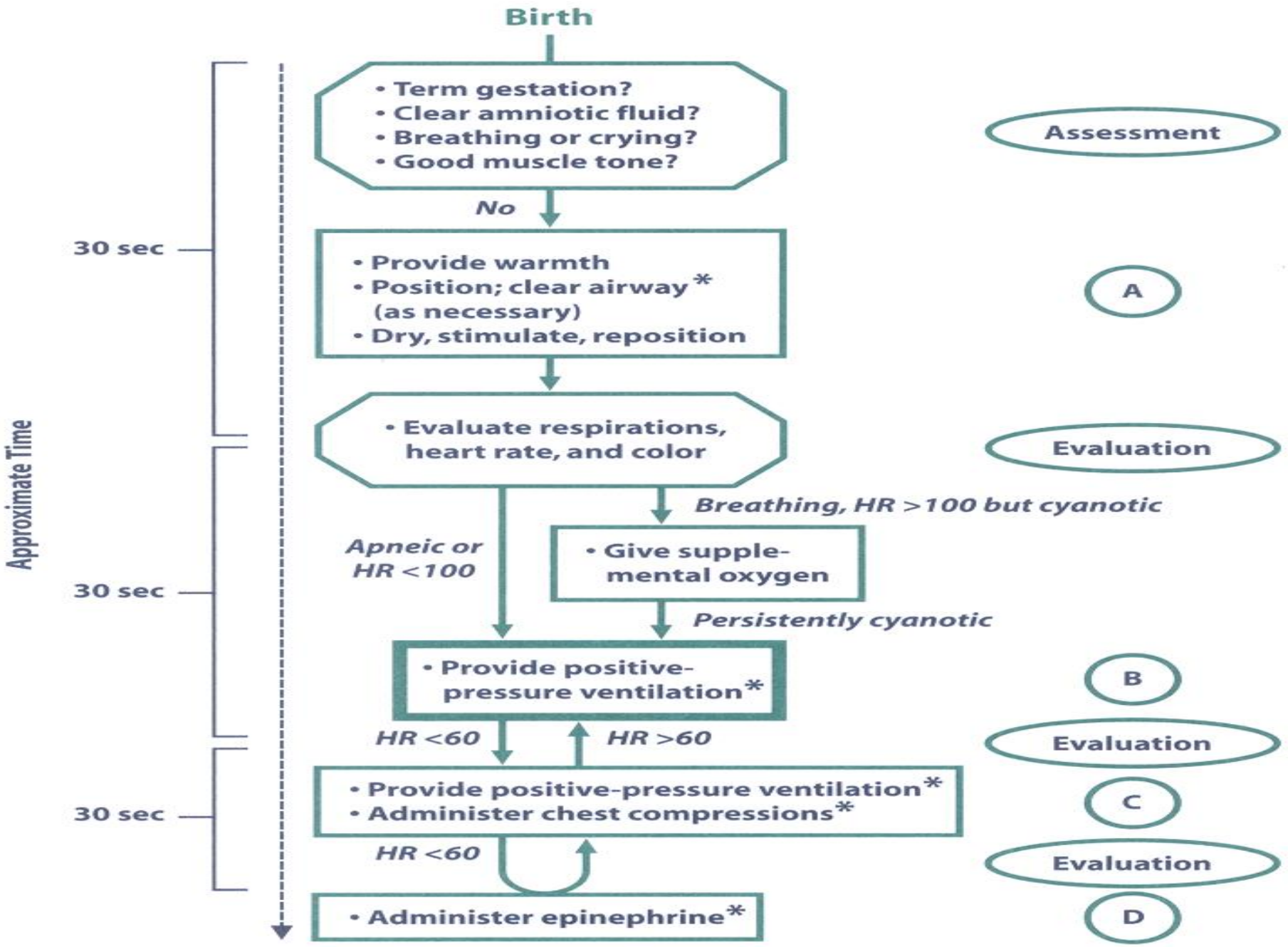
Dr; hekma Saad farghaly

The newborn: is the live born infant in the first 28 days of life adjusting for extra uterine life.

Neonatal resuscitation

- Most infants do not require any resuscitation. Shortly after birth, the baby will gasp or cry, establish normal breathing and become pink. The baby can be handed directly to his or her mother.

- When delivery is complicated by perinatal conditions leading to asphyxia, resuscitation aims to reverse hypoxia, hypercarbia, and acidosis. The survival and outcome of distressed newborns depend on timely and effective intervention in the first few minutes after birth. Newborn infant who does not establish normal respiration directly will need to be transferred to a resuscitation table for further assessment.



* Endotracheal intubation may be considered at several steps.

Preterm infants

Definition: newborn infant born before 37 weeks gestation.

Common disorders of preterm infants

1- Respiratory problems: respiratory distress syndrome and apneic attacks.

2- Alimentary tract problems: abdominal distension, regurgitation of gastric contents.

3- Hyperbilirubinemia. 4- Hypothermia.

5- Hypoglycemia. 6- Hypocalcaemia.

7- Anemia. 8- Edema.

9- Circulatory instability.

10- Infection.

11- Cerebral anoxia.

Feeding of preterm infants:

feeding may be parenteral or enteral

Parenteral nutrition

- Most very small immature babies need their fluid and calorie requirements intravenously for at least the first 48 hours.
- Intravascular administration of 5-10% dextrose solution is initially given for babies under 1500 gm, those with respiratory distress and all ill babies.
- More prolonged maintenance of nutrition intravascularly is made possible by giving a balanced mixture of amino acids, glucose, fats and electrolytes in what is called total parenteral nutrition ”hyperalimentation”.

Enteral nutrition

- ***a- Breast milk:*** breast milk from infant's mother is more appropriate. It has the merit of being well tolerated, better absorbed in the gut and provides some protection against infection.
- ***b- Formula milk:*** specialized preterm formula adapted for the nutritional needs of low birth weight infants is available and can be used if breast milk is not available or insufficient.

Vitamin and iron supplementation

- ***Vitamin D:*** preterm infant has an increased liability to develop rickets after about 2 months of age and therefore breast fed babies should receive vitamin D (1000 IU/day).
- ***Vitamin A***: (1000 IU/kg/day)
- ***Vitamin C:*** (50 mg/ day)
- ***Vitamin E:*** (10 mg daily): may be given to protect against hemolytic anemia.
- ***Folic acid:*** (100 μg daily) should be given to very low birth weight babies to prevent a later macrocytic anemia.
- ***Iron:*** (2.5 mg/kg daily) is given from 4-6 weeks of age onwards to prevent anemia of prematurity.

Post-term infant

Definition: *the newborn's gestation exceeds 42 weeks.*

Neonatal disorders

Apnea

Apnea: means cessation of breathing

Significant apnea: means cessation of breathing for longer than 20 sec, or any duration if accompanied by cyanosis and bradycardia.

Apnea is a common problem in preterm infants that may be due to idiopathic apnea of prematurity or an associated illness

Treatment:

- Treatment of the cause
- Infants at risk for apnea should be monitored with apnea monitor.
- Gentle ***cutaneous stimulation*** is often adequate therapy for neonatal infants with mild and intermittent episodes.
- Infants with recurrent and prolonged apnea require immediate ***bag and mask ventilation.***
- ***Oxygen*** should be administered carefully to treat hypoxia.
- Recurrent apnea of prematurity not due to precipitating identifiable cause may be treated with ***theophylline*** or *caffeine*.
- *Oxygen therapy by* high-flow nasal cannula *or assisted ventilation by nasal continuous positive airway pressure* are effective therapies for mixed or obstructive apnea.

Cyanosis
Central cyanosis

Causes:

A- Respiratory insufficiency

B- Congenital cyanotic heart disease

I- Respiratory distress syndrome (RDS) of the newborn *(Hyaline membrane disease)*

To maintain alveolar stability, surfactant reduces the surface tension and prevents collapse of small air spaces at end of expiration. The primary cause of respiratory distress syndrome (hyaline membrane disease) is inadequate pulmonary surfactant with consequent diffuse alveolar atelectasis, edema, and cell injury.

Pneumonia

Types

Congenital pneumonia: early onset of respiratory distress within 3-6 hours is observed. Clinical findings may be similar to hyaline membrane disease. There may be temperature instability and apneic spells.

Acquired pneumonia: late onset of respiratory distress (at any time of neonatal period) commonly follows aspiration, mechanical ventilation or septicemia.

Specific treatment

- Surfactant replacement therapy (delivered through endotracheal tube) for cases with hyaline membrane disease.
- Specific antibiotics for cases with pneumonia to eradicate infection.
- Fluid therapy aiming at adequate hydration and tissue perfusion.
- Naloxone as an antidote is given to patients with CNS depression due to maternal morphine overdose.

Anemia in the newborn infant

Cord blood hemoglobin is 16.8 g/dL. A "physiologic" decrease in hemoglobin is noticed at 8-12 weeks. Anemia at birth is manifested by pallor, congestive heart failure or shock. It is caused by hemolytic disease of the newborn, but may be the result of traumatic blood loss.

Treatment

The baby who has repeated apnea and bradycardia and hemoglobin <12 gm/dL may benefit from blood transfusion to improve oxygen delivery. Packed red blood cell transfusion is given slowly.

Neonatal hypoglycemia

Hypoglycemia is usually defined as a blood glucose value of < 40 mg/dL.

Pathogenesis

Hypoglycemia is due to increased glucose utilization (hyperinsulinism) and /or decreased production /stores of glucose.

Risk factors of neonatal hypoglycemia

1- Infants of diabetic mothers (IDM).
2- Large for gestational age infants (LGA).
3- Small for gestational age infants (SGA).
4- Preterm and post-term infants.
5- Sick or stressed infants e.g. respiratory distress syndrome or hypothermia.
6- Fasting infants.
7- Infants with polycythemia.
8- Erythroblastosis fetalis.
9- Maternal drugs e.g. steroids, beta-sympathomimetics and beta blockers.

Management

Stable near-term or term infants at risk for hypoglycemia should be fed early soon after birth. Breast milk or 5% glucose should be provided if the baby is able to tolerate nipple or nasogastric tube feedings. These infants should be monitored for blood glucose level until they are taking full feedings and have three normal prefeeding readings of blood glucose (above 40 mg/dL). If the infant at risk for hypoglycemia is unable to tolerate nipple or tube feedings, maintenance IV therapy with 10% glucose should be initiated and glucose level is monitored.

Neonatal convulsions

Etiology

1- Cerebral anoxia.

2- Intracranial hemorrhage especially intraventricular hemorrhage that is more common in premature babies.

3- Intracranial infections.

Treatment

1- Supportive care: ventilation, nutrition, normal body temperature, serum electrolyte and blood pressure monitoring.

2- Treatment of the underlying cause.

3- Anticonvulsants:

a- IV diazepam (0.1-0.3mg/kg/ dose) for immediate stoppage of convulsions.

b- Phenobarbitone: loading dose (20 mg/kg IV over 15 minutes) may be used after 1 hour if convulsions persist. Maintenance dose 5 mg/kg/ day orally or IV as a single or divided doses.

c- Phenytoin: is given when there is no response to phenobarbitone. Loading dose 15-20 mg/kg in saline not greater than 1 mg/kg / minute. Maintenance dose 5 mg/kg/day IV in 2-3 divided doses.